

SHAILZA JOLLY

📍 Berlin, Germany

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PROFILE

ML Scientist/Engineer with a Ph.D. in CS and 6+ years spanning academic research and industry. Specializes in LLMs, Generative AI, and scalable data pipelines, with hands-on depth in modern LLM architectures, efficiency techniques (KV-cache, speculative decoding, FlashAttention), and agentic system design. Pairs a track record of top-tier publications (AAAI, NAACL, EMNLP) with shipping production systems and translating technical work into clear business outcomes. I'm exploring what's next at the intersection of research, engineering, and product. I am open to exceptional roles where I can take hard problems from research all the way to real users.

SKILLS

- **ML/GenAI:** LLMs, NLU/NLG, multimodal learning, synthetic data, model evaluation.
- **LLM systems:** Agentic workflows (MCP, LangGraph/LangSmith), RAG/grounding concepts, experiment design & offline evaluation.
- **Data & scale:** Python, PyTorch, PySpark; large-scale text processing/data quality.
- **Search and retrieval:** embedding models (e.g., Sentence Transformers), reranking, vector search.
- **Platforms:** AWS; production-grade coding, documentation, and cross-functional delivery.

WORK EXPERIENCE

Applied AI & LLM Systems

July 2025 - Present

Independent

Berlin, Germany

- Working hands-on at the intersection of science, engineering, and product, building and iterating on a series of prototypes across agentic systems, retrieval, and LLM evaluation while exploring directions worth committing to.
- Refreshed modern LLM foundations (Stanford) and went deep on architecture and efficiency internals. One example prototype: Sift, an agentic RAG system that returns grounded, citable answers over a corpus of technical PDFs.

Flinn.ai

January 2025 - June 2025

Senior AI Engineer

Berlin, Germany

- Led development of AI-driven product features including data extraction from medical research papers and multilingual complaint monitoring.
- Partnered with product and backend teams to scope problems, define success metrics, and deliver end-to-end features within an agile product lifecycle.

Parental Leave (Maternity)

December 2023 - December 2024

- Parental leave; stayed engaged with research literature and maintained technical skills.

Amazon AI

July 2022 - November 2023

Research Scientist

Berlin, Germany

- Led development of a scalable noise-removal/data-quality pipeline processing billions of tokens to improve training data for LLMs.
- Wrote research plans and technical documentation; presented results and recommendations to senior science/engineering leadership.
- Mentored Master's/Ph.D. interns and collaborated on research deliverables.

German Research Center for Artificial Intelligence (DFKI)

March 2019 - June 2022

Machine Learning Scientist

Berlin, Germany

- Delivered ML research and engineering (coding, experimentation, publications) in BMBF-funded projects: *XAINES* (Explainable AI) and *DeFuseNN* (vision-language systems).
- Supervised interns and BSc/MSc students; provided technical mentorship and project guidance.

NVIDIA Research*Machine Learning Scientist Intern*

May - August 2021

Santa Clara, United States

- Built a document understanding pipeline combining table/cell detection, tabular structure retrieval, and OCR for financial documents.

Amazon Alexa*Applied Scientist Intern*

August 2019 - December 2019

Aachen, Germany

- Developed a method for generating diverse synthetic training data to improve intent classification and slot labeling for task-oriented NLU.

SAP AI Research*Research Intern / Master's Thesis*

May 2018 - February 2019

Berlin, Germany

- Designed and implemented an evaluation metric for Visual Question Answering (VQA) models.

EDUCATION

TU Kaiserslautern, Germany*Ph.D. in Computer Science*

March 2019 - June 2022

Building Natural Language Generation and Understanding Systems in Data-Constrained Settings; work on VQA, interpretability, and conversational AI.

Overall grade: **“Sehr Gut” 1.0** (1.0 is highest on 1.0–5.0 scale)

University of Copenhagen, Denmark*Visiting Researcher*

January 2021 - March 2021

Developed an unsupervised post-editing algorithm to generate fluent fact-checking explanations.

TU Kaiserslautern, Germany*M.Sc. in Computer Science — Minor in Economics*

October 2016 - November 2018

Overall grade: **“Sehr Gut” 1.5**

Kyushu University, Japan*Semester Abroad*

November 2017 - February 2018

Explainable AI to analyze behavior of deep CNN architectures for image recognition.

Guru Nanak Dev Engineering College, India*B.Tech in Computer Science & Engineering*

August 2012 - July 2016

Overall grade: **First Division with Distinction**

SELECTED PUBLICATIONS

- **Jolly, S.**, Zhang, Z. X., Dengel, A., & Mou, L. (2022). Search and learn: Improving semantic coverage for data-to-text generation. *AAAI*.
- **Jolly, S.**, Pezzelle, S., & Nabi, M. (2021). EaSe: A diagnostic tool for VQA based on answer diversity. *NAACL-HLT*.
- **Jolly, S.**, & Kapoor, S. (2020). Can pre-training help VQA with lexical variations? *Findings of EMNLP*.
- **Jolly, S.**, Falke, T., Tirkaz, C., & Sorokin, D. (2020). Data-efficient paraphrase generation to bootstrap intent classification and slot labeling. *COLING (Industry Track)*.
- **Jolly, S.**, Iwana, B. K., Kuroki, R., & Uchida, S. (2018). How do convolutional neural networks learn design? *ICPR. (Best Student Paper)*

AWARDS AND HONORS

- Received **AAAI-22 Scholarship** by Hitachi to attend AAAI 2022 (2022).
- Awarded **AI Newcomer 2021 Award** (German Informatics Society and BMBF, Germany) (2021).
- Received **EU-Cost STSM Grant** to work on Multi3generation project at CopeNLU (2020).
- **Best Student Paper Award** at ICPR (2018).